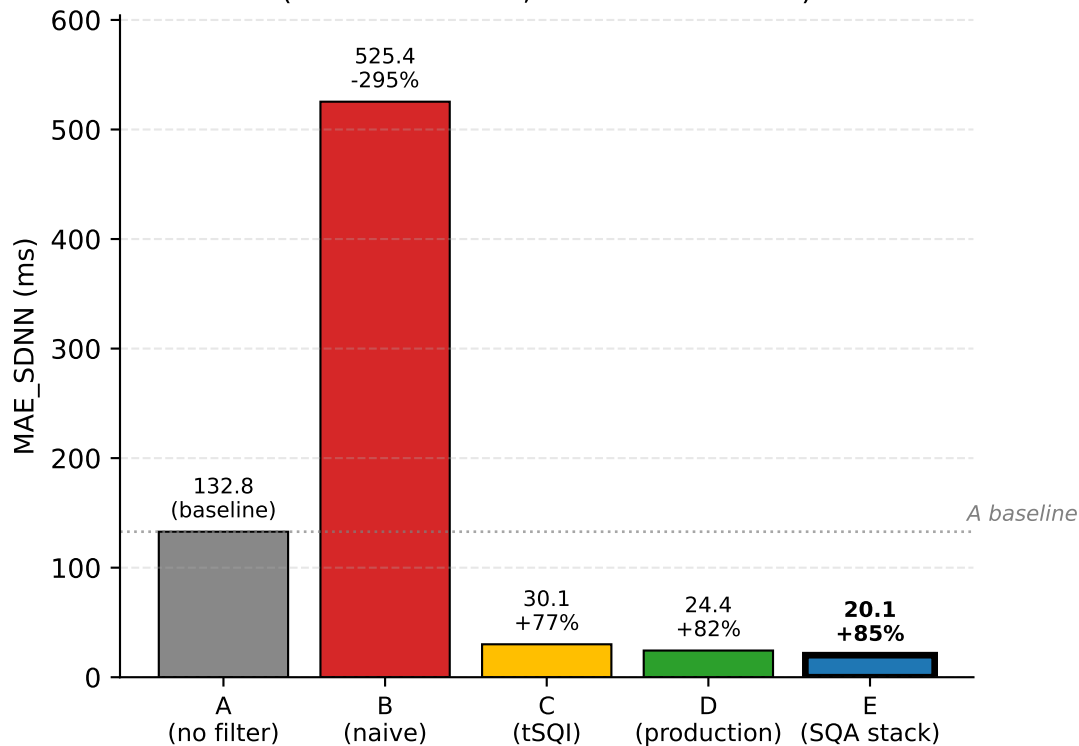
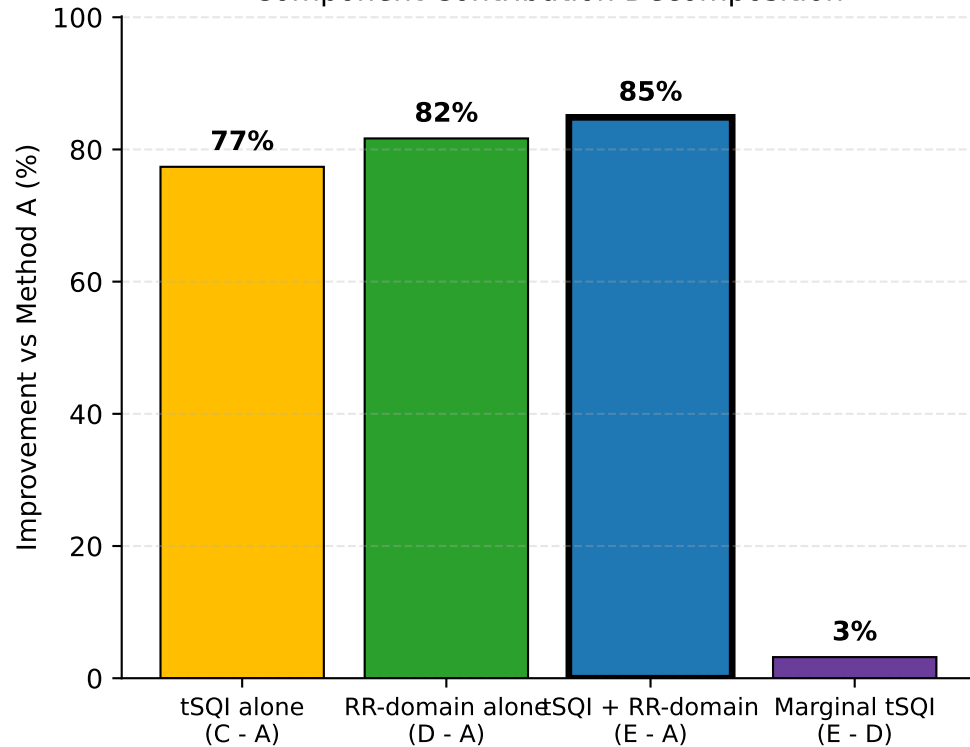


HRV SDNN MAE - 5-Method Comparison
(PPG-DaLiA unfit, matched N=1871)



Component Contribution Decomposition



Method A: no HRV filter (baseline). Method B: naive `np.diff(kept_peaks)` after tSQL hard threshold (Day 7 strawman, gap-spanning RR bug). Method C: HeartPy van Gent 2019 vectorized mask (`accepted[:-1] & accepted[1:]`) per `heartpy/analysis.py:293`. Method D: production HeartPy 2-stage RR rejection (Task Force 1996 + +/-20% median quotient), mirrors Backend code/main.py: `_reject_rr_outliers`. Method E: combined C + Stage 2 quotient. Matched comparison N=1871 (windows where all 5 methods produced valid output). Per-method-success N=7061 gives MAE_A 146.43 ms for full-population reference.